

AI Project Management Ecosystem — Case Study Definition

Version: 1.0

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1. Overview

The **AI Project Management Ecosystem** is a web-based platform that augments traditional project management with a multi-agent artificial intelligence ecosystem. It integrates with the Odoo 19 platform and provides project managers and team members with a natural-language chat interface, interactive dashboards, and automated reporting.

The system is designed for small to medium-sized organizations that manage multiple software projects in parallel. It addresses two main needs:

- ****Reduce the friction of routine project management**** — creating projects, adding tasks, moving tasks between stages, and deleting obsolete items — by allowing users to perform these operations through natural language.
- ****Automate periodic reporting**** — weekly status reports summarizing all projects, tasks, risks, and progress — delivering them to project managers without manual effort.

The platform combines a **multi-agent AI orchestrator** (which routes user requests to specialized AI agents) with **role-based access control** (differentiating project managers from team members) and ****persistent chat sessions**** (allowing users to resume conversations across days and switch between large language model providers).

2. Purpose and Scope

The system SHALL allow users to:

- Interact with the ecosystem through a **natural language chat interface**.
- Maintain **multiple persistent chat sessions** per user, each preserving the full conversation history.
- **Select the AI model provider** (Groq, OpenAI, Anthropic) for each chat session.
- Perform **read operations**: list projects, list tasks, list stages, analyze risks, prioritize tasks, and generate project summaries.
- Perform **write operations** (project managers only): create projects, add tasks, create stages, move tasks between stages, delete tasks, delete stages, and delete projects.
- Visualize **global and per-project dashboards** with interactive charts.
- Export **PDF reports and CSV files** per project.
- **Automatically generate and deliver** a weekly ecosystem report.

The system integrates with the **Odoo 19 platform** as the source of truth for project and task data, and uses a separate **AI microservice** (built on the CrewAI framework) to orchestrate the specialized AI agents.

3. Stakeholders

The following stakeholders have an interest in the AI Project Management Ecosystem:

- **Project Managers** — primary users who plan, execute and monitor projects.

They require full write access to the ecosystem, including the ability to create, modify and delete projects, tasks and stages.

- **Team Members** — users who contribute to projects and need visibility over their assigned tasks, stages and risks. They require read-only access to the ecosystem and cannot modify any data.
- **Product Managers** — users focused on the product backlog and prioritization. They consume dashboards and prioritize tasks but do not manage day-to-day execution.

- ****System Administrators**** — users responsible for configuring the ecosystem, managing users, monitoring the AI microservice, and configuring automated reports.
- ****The Organization**** — the entity that owns the projects and benefits from reduced management overhead and automated reporting.

4. Functional Description

4.1 Project Management

The system provides functionality to:

- Create a new project with a name and an optional list of initial tasks.
- List all existing projects, showing each project's name and task count.
- Delete an existing project together with all its associated tasks.
- View a per-project dashboard with tasks, stages, contributors, and

interactive charts.

4.2 Task Management

The system provides functionality to:

- Add a new task to an existing project, initially placed in a "no stage" state.
- List all tasks of a project, showing each task's name and current stage.
- Move one or more tasks to a target stage within the same project.
- Delete one or more tasks from a project.
- Prioritize tasks, placing tasks without a stage first.

4.3 Stage Management

The system provides functionality to:

- Create a new stage inside a project, with an associated ordering sequence.
- List all stages of a project.
- Delete a stage, subject to the constraint that the stage must not contain

any task.

4.4 Multi-Agent AI Chat

The system provides functionality to:

- Open a natural-language chat interface and interact with a specialized

AI agent.

- Create multiple chat sessions per user, each with a distinct name and

independent conversation history.

- Switch between available large language model providers per session.
- Route each user message to the most appropriate specialized agent:

a Project Manager agent, a Team Member agent, a Reporting agent, or a Diagram agent.

- Analyze project risks automatically when requested, based on the number of tasks without an assigned stage.

4.5 Dashboards and Reporting

The system provides functionality to:

- Present a global dashboard with summary indicators (total projects, total tasks, total stages, average lead time, tasks per day).
- Display interactive charts: tasks by stage (donut), tasks by creator (bar), and detailed breakdowns by stage and by creator.
- Present a per-project sub-dashboard with tasks, stages, contributors and two charts (tasks by stage and tasks by assignee).

4.6 Automated Weekly Reports

The system provides functionality to:

- Automatically generate a weekly status report at a scheduled interval (every one week).
- Include in the report the total number of projects, total number of tasks, and a per-project breakdown with task count, tasks without stage, and estimated risk level.
- Deliver the report through the platform's internal email system.

4.7 Role-Based Access Control

The system provides functionality to:

- Identify whether the current user belongs to the "Project Manager" group

(full access) or the "Team Member" group (read-only access).

- Restrict all write operations (create, add, delete, move, update) to users

in the "Project Manager" group.

- Deny write requests issued by team members with an explicit refusal

message.

- Display a role badge and a role-specific home page for each user.

5. Business Rules

The following business rules govern the system's behavior:

- A **stage** cannot be deleted while it contains at least one task.
- A **project** deletion cascades to all its tasks.
- A **task** is initially created without an assigned stage and can be moved to any stage of the same project.
- All write operations are **idempotent**: repeating the same command yields the same result without creating duplicates.
- A **chat session** is scoped to a single user and cannot be accessed by another user.
- A **team member** cannot perform any write operation; such attempts are rejected with an explicit denial message.
- The **weekly report** is generated once per week and does not repeat on failure — a failed delivery is logged but does not block the schedule.
- Each project has at most one **personal stage set** that is shared across the ecosystem.

6. Non-Functional Aspects

- **Usability:** The chat interface responds in natural language and must be

usable by non-technical users.

- **Performance:** The system should return a chat response within fifteen seconds for ninety percent of requests.
- **Reliability:** The system must handle AI provider rate limits gracefully by retrying failed requests.
- **Security:** All write operations are protected by role-based access control and executed under an authenticated Odoo user.
- **Extensibility:** The system must allow adding new AI providers, new specialized agents, and new tools without modifying the existing codebase.
- **Internationalization:** All user-facing strings, agent instructions, and documentation must be written in English.
- **Attribution:** Every page of the ecosystem must display the author and institutional copyright.

7. Out of Scope

The following aspects are explicitly out of scope for version 1.0 of the case study definition:

- Real-time collaboration features (simultaneous editing of the same task).
- Advanced access-control beyond the two roles (Project Manager, Team Member).
- Integration with external issue trackers (Jira, GitHub Issues, etc.).
- Authentication through identity providers (SSO, OAuth).
- Mobile applications.
- Fine-grained audit logs per field.

End of case study definition.